Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase II Trial of Tisagenlecleucel in First-line High-risk (HR) Pediatric and Young Adult Patients With B-cell Acute Lymphoblastic Leukemia (B-ALL) Who Are Minimal Residual Disease (MRD) Positive at the End of Consolidation (EOC) Therapy **Short Title/Acronym:** CASSIOPEIA **Protocol Number:** CCTL019G2201J **NCT Number:** NCT03876769 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Tisagenlecleucel (CTL019 / Kymriah) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of tisagenlecleucel therapy as measured by the 5-year disease-free survival (DFS) rate by investigator assessment. **Secondary Objectives:** * To assess the proportion of subjects who are disease-free without allogeneic Stem Cell Transplantation (SCT) at 1 year. * To assess Overall Survival (OS). * To assess the proportion of subjects achieving MRD negative Complete Remission (CR) or CR with incomplete blood count recovery (CRi) at month 3 post-tisagenlecleucel infusion. * To assess the tisagenlecleucel manufacturing success rate.

2.2 Background & Rationale To determine the efficacy and safety of tisagenlecleucel in de novo High-Risk (HR) pediatric and young adult B-ALL patients. These patients have received first-line treatment and remain End of Consolidation (EOC) Minimal Residual Disease (MRD) positive, putting them at high risk for relapse, thereby necessitating advanced cellular therapy interventions.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: Approximately 122 participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 30-Jun-2019 **Estimated Primary Completion:** Approximately 8 years after First Patient First Treatment (FPFT)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 1 to 25 years at the time of screening. 2. CD19 expressing B-cell Acute Lymphoblastic Leukemia. 3. De novo NCI HR B-ALL who received first-line treatment and are MRD $\ge 0.01\%$ at EOC. 4. Lansky (age < 16 years) or Karnofsky (age ≥ 16 years) performance status $\ge 60\%$ at screening.

4.2 Exclusion Criteria 1. M2 or M3 marrow or persistent extramedullary disease at the completion of first-line consolidation therapy. 2. Philadelphia chromosome positive ALL or prior tyrosine kinase inhibitor therapy. 3. Concomitant genetic syndromes associated with bone marrow failure states (e.g., Fanconi anemia, Kostmann syndrome).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single infusion of tisagenlecleucel **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Single infusion with short-term acute monitoring and long-term survival follow-up (up to 8 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard bridging or lymphodepleting chemotherapy prior to infusion. **Prohibited:** Prior gene or engineered T-cell therapy, or any prior anti-CD19 therapy.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Pre-treatment	Informed Consent, Leukapheresis, MRD Status confirmation
Baseline	Day 0	Tisagenlecleucel Infusion
Interim	First month & Day 29	High-frequency efficacy and safety assessments, CRS monitoring
End of Study	Up to 8 Years	Q3M for Year 1; Q6M for Year 2; then yearly. Long-term Lentiviral safety follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 5-Year Disease-Free Survival (DFS) rate. DFS is defined as the time from the date of infusion to the date of the first documented morphological relapse, secondary malignancy, or death due to any cause. **Measurement Method:** Investigator assessment.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) rate. * Proportion of subjects achieving MRD negative CR or CRi at Month 3. * Proportion of subjects maintaining complete remission with persistent B-cell aplasia over time.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at study visits and as clinically indicated, with special focus on Cytokine Release Syndrome (CRS) and neurotoxicity. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines. Long-term follow-up for lentiviral vector safety will continue under a separate protocol per health authority guidelines. **Data Monitoring Committee (DMC):** External independent data monitoring committee utilized per sponsor guidelines.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses and Safety Set for adverse events. **Sample Size Justification:** Powered to adequately evaluate 5-year DFS improvement versus historical standard-of-care controls. **Handling of Missing Data:** Standard time-to-event methodologies (Kaplan-Meier survival curves); censoring for new anticancer therapy depending on the objective measured.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict onsite monitoring during the pre-treatment and acute infusion phases, with long-term follow-up data collection. **Audit Trail:** Robust electronic Case Report Form (eCRF) utilizing central laboratory analysis for flow cytometry (MRD assessment).

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11. Study Identifiers & Governance

Primary ID: CCTL019G2201J **Registry Number:** NCT03876769 **Sponsor/ Lead Organization:** Novartis Pharmaceuticals **Study Title:** CASSIOPEIA **Official Regulatory Title:** A Phase II Trial of Tisagenlecleucel in First-line High-risk (HR) Pediatric and Young Adult Patients With B-cell Acute Lymphoblastic Leukemia (B-ALL) Who Are Minimal Residual Disease (MRD) Positive at the End of Consolidation (EOC) Therapy

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12. Clinical Framework (B-ALL Specific)

Primary Condition: B-cell Acute Lymphoblastic Leukemia (B-ALL) **Diagnosis Threshold:** MRD $\ge 0.01\%$ at End of Consolidation (EOC) **Medical Methodology:** CAR-T Cell Therapy (Tisagenlecleucel)

Optimization Target: MRD-negative Complete Remission and long-term Disease-Free Survival

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13. Study Procedures & Methodology

Management Pathway: Single-infusion CAR-T cell therapy administered after first-line induction/consolidation **Education Protocol (HCP):** Specialized site certification and training for CAR-T administration and CRS management **Education Protocol (Patient):** Comprehensive pre-infusion counseling and strict post-infusion monitoring requirements **Quality Audit Mechanism:** Tisagenlecleucel manufacturing success rate assessment

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 8 years post-infusion **Onsite Visit Cadence:** Frequent monitoring Month 1; Day 29; Every 3 months (Year 1); Every 6 months (Year 2); Yearly thereafter **Remote Monitoring Frequency:** As required for long-term survival follow-up **Checklist Review Interval:** Corresponding to primary efficacy measurement cycles (Day 29, Month 3, Year 1, Year 5)

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **| *[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |